

①

Start



select message and Key of equal length

convert both to ASCII



convert this into binary



for each letter, perform an $\oplus$ operation
each binary message letter and its
corresponding key letter (in binary)



convert the results to hexadecimal,
this is your encrypted message

to decrypt, perform XOR operation

on binary form of encryption message and the
binary key.

②

The biggest problem I faced while completing this activity is keeping track of long binary numbers and making sure each ~~step~~ was done properly. Working with binary numbers by hand is not very enjoyable...

③ Encryption

secret word = "wave";

$$w = 119 = 01110111$$

$$a = 097 = 01100001$$

$$v = 118 = 01110110$$

$$e = 101 = 01100101$$

Key = "loud":

$$l = 108 = 01101100$$

$$o = 111 = 01101111$$

$$u = 117 = 01110101$$

$$d = 100 = 01100100$$

$$\begin{array}{r} w \oplus l = 01110111 \\ \quad \quad 01101100 \oplus \end{array} = 00011011 = 1B$$

$$\begin{array}{r} a \oplus o = 01100001 \\ \quad \quad 01101111 \oplus \end{array} = 00001110 = 0E$$

$$\begin{array}{r} v \oplus u = 01110110 \\ \quad \quad 01110101 \oplus \end{array} = 00000011 = 03$$

$$\begin{array}{r} e \oplus d = 01100101 \\ \quad \quad 01100100 \oplus \end{array} = 00000001 = 01$$

Encrypted output = 1B 0E 03 01

④

Decryption

$$\begin{array}{r} 00011011 \\ 01101100 \end{array} \oplus = 01110111 = w$$

$$\begin{array}{r} 00001110 \\ 01101111 \end{array} \oplus = 01100001 = a$$

$$\begin{array}{r} 00000011 \\ 01110101 \end{array} \oplus = 01110110 = v$$

$$\begin{array}{r} 00000001 \\ 01100100 \end{array} \oplus = 01100101 = e$$